

Capacitor Charging

Pina cammisano

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1 Introduction

This document introduces the charging behavior of a resistor-capacitor (RC) network. When a capacitor C is charged through a series resistor R from a constant supply V_0 , the capacitor voltage rises according to

$$V(t) = V_0 \left(1 - e^{-t/RC}\right)$$

where the product RC has units of second and is called the time constant, $T = RC$